



FRENCH-GERMAN ADSORPTION INITIATIVE

SCIENTIFIC PROGRAM

**Duisburg,
Germany 24-26
March 2026**

Welcome

The 2nd Adsorption Conference organized by the “French-German Adsorption Initiative” will be held in Duisburg (Germany) between March 24th and 26th, 2026. The conference is supported by the German Adsorption Group (DECHEMA) and the French “Association Francaise de l’Adsorption (AFA)”. The aim of the conference is to bring together and foster collaboration between scientists from these two communities. However, we see this joint initiative as a platform that could be extended to all European countries in the very near future. Therefore, we are open for presentations from France, Germany and neighboring countries.

Adsorption phenomena play a crucial role in many processes, with applications ranging from catalysis to treatment of waste streams. Current research ranges from the development of innovative adsorbent materials to using molecular modeling tools to predict adsorption behavior. The scope of adsorption research is also expanding into new fields, including energy systems, environmental protection, and the life sciences.

As we want to cover the whole bandwidth from basic research to applied technology, we invite people both from academia and industry to present their topics. This creates an excellent environment for the exchange of ideas and knowledge between scientists and practitioners in the field.

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Dieter Bathen (Chairman),
Benjamin Claessens (Junior-Chairman)
& Christian Bläker (Junior -Chairman)

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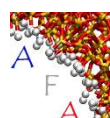
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P.01: YoungFluidSeps Early Career Fluid Dynamics and Separation Engineers

G. Chiari | Muenster University of Applied Sciences (DE)

P.02: Adsorption measurements and modeling in the context of high-pressure natural gas dehydration

A. Grim | Université de Pau et des Pays de l'Adour / TotalEnergies (FR)

P.03: Co-Adsorption of Water and Carbon Dioxide on a Commercial Material for Direct Air Capture (DAC)

A. Kretzschmar | Forschungszentrum Jülich GmbH (DE)

P.04: Temperature and moisture dependent adsorption kinetics of Polyethylenimine coated carbon nanofibers

C. Jung | Forschungszentrum Jülich GmbH (DE)

P.05: Model-Based Optimization of Phosphate Recovery From Municipal Wastewater Using Iron Hydroxide Fixed-Bed Adsorption

C. T. Zipp | RPTU University Kaiserslautern-Landau (DE)

P.06: Lattice Boltzmann Method for the adsorption and transport of complex pollutants in bimodal porous media

D. Bauer | IFPEN (FR)

P.07: Isothermal vs. adiabatic process investigations of CO₂/H₂O co-adsorption effects on amine-based adsorbents under flue gas conditions

D. Neu | OTH Regensburg / TU München (DE)

P.08: Radon Adsorption in Cation-Exchanged Ferrierite: A Combined molecular simulation – experimental study

E. B. Zerifi | Université de Haute-Alsace, IS2M (FR)



P.09: CO₂ Capture using porous silica-based materials at multiscale porosity

G. Rodriguez Almaquer | Université de Pau et des Pays de l'Adour (FR)

P.10: VPSA Process For Capturing CO₂ From Industrial Flue Gas at Post-Combustion Conditions Using Experimental and Simulation Procedures

H. Al Awaad | University of Mons (FR)

P.11: New adsorption database containing 1000+ experimental datasets for vapor and gas adsorption on zeolites and other porous solids

I. Bezverkhyy | CNRS - Université Bourgogne Europe (FR)

P.12: Advanced characterization of nanoporous materials using a novel in-situ vapor sorption calorimetry method

J. Dümmler | FAU Universität Erlangen-Nürnberg (DE)

P.13: TSA Plant Design: Optimizing Regeneration Conditions

J. Steinhaus | Linde GmbH (DE)

P.14: Combining Induction Heating and 2-D Modeling to Investigate Heat and Mass Transfer Phenomena in Fixed-Bed Adsorption

M. Gholami | Vrije Universiteit Brussel (BE)

P.15: Effect of the Regeneration Method on the Characterization of Platinum Nanoparticles using H₂ Chemisorption

N. Janik | Forschungszentrum Jülich (DE)

P.16: Adsorption of Np(V) and U(VI) onto zirconia (ZrO₂): a batch, spectroscopic, and modeling study

N. Jordan | Helmholtz-Zentrum Dresden-Rossendorf e.V., Institute of Resource Ecology (DE)



P.17: Selective radon adsorption on carbon materials

P. S. Ortega-Rivera | Université de Lorraine (FR)

P.18: Investigation of Physics-informed Neural Networks for Surrogate Modeling of Chromatographic Systems

R. Reifsteck | RWTH Aachen University; Chair for Fluid Process Engineering (DE)

P.19: Binary CO/N₂ separation on Ba-ETS-4 and comparing with 5A and 13X zeolites

S. Saadat | Vrije Universiteit Brussel (BE)

P.20: Graphene-Based Metal Oxide Composites for the Adsorptive Elimination of Crystal Violet

S. Arif | Department of Chemistry, COMSATS University Islamabad (PAK)

P.21: Sustainable Adsorptive Dye Removal Using Walnut Septum-Derived Activated Carbon

S. Arif | Department of Chemistry, COMSATS University Islamabad (PAK)

P.22: Stability of Adsorbents for Direct Air Capture (DAC): Experimental and Modeling Approach

S. Fakhraddin | Université de Liège (FR)

P.23: An analytical design model for biogas upgrading VPSA plants

T. Welter | Frankfurt University of Applied Sciences (DE)

P.24: Methanol from Air – The DryHy Project – Water-positive Methanol Generation by Utilizing Amine-based Direct Air Capture

V. Selmert | Forschungszentrum Jülich GmbH (DE)

P.25: Coal tar distillers in transition: developing new products for a sustainable, carbon-neutral economy (DISTINCTION)

Y. Hammi | CNRS (FR)



[P.26:](#) Adsorption of acetone on silica-alumina gels

[J. Beerlage](#) | University of Duisburg-Essen (DE)

[P.27:](#) Experimental investigation of the degradation of amine-functionalised adsorbents during direct air capture processes

[C. Schwarz](#) | RWTH Aachen University (DE)

[P.28:](#) Beyond Vycor Glass – New Reference Materials for Adsorption Experiments

[P. Mangen](#) | Leipzig University (DE)

[P.29:](#) Beyond forever sorbents: Capturing the impact of sorbent degradation in adsorption based direct air carbon capture and storage systems

[A. Snackers](#) | RWTH Aachen University (DE)

[P.30:](#) PSA-PINN: An Integrated Platform for the Collaborative Design, Optimization and Control of Pressure Swing Adsorption Processes and Adsorbents

[W. Guo](#) | Tianjin University (CHN)

[P.31:](#) Estimation of heat and mass transfer during desorption for direct air capture using the infrared-large-temperature-jump method

[T. Aziz](#) | RWTH Aachen University (DE)

[P.32:](#) A decoupled DR-PSA process for upgrading low- quality natural gas using optimized ETS-4

[Y. Wang](#) | Tianjin University (CHN)

[P.33:](#) Adsorption Kinetics Model of Hydrogen on Graphite

[J.-M. Simon](#) | ICB CNRS Université Bourgogne Europe (FR)



P.34: Kinetics of CO₂ Adsorption and Deformation in Microporous Carbon

A. Kolesnikow | Leipzig University (DE)

P.35: Study of adsorption/desorption of carbon dioxide using microwave heating (MSA) on modified Metal Organic Frameworks (MOF's)/ Graphene Oxide (GO) composites

A. Houzé | University of Mons (BE)

P.36: Textile Dye Desorption to Enable Activated Carbon Reuse

A.-M. Heidrich | Technische Universität Braunschweig (DE)

P.37: Confined Geometry Effects on the Supercritical Adsorption Behavior of Hydrogen

J. Xu | Friedrich-Alexander-Universität Erlangen-Nürnberg (DE)

P.38: From Powder to Pellet: Impact of Binders on the Structure, Functionality and Robustness of Carbon Nanofiber Molecular Sieves

M. Pagel | Forschungszentrum Jülich (DE)

P.39: Comparative Physisorption Performance of MOR-Type Zeolites and KAUST-7 for Direct Air Capture under Humid Conditions

M. Bonneau | IRCELYON (FR)

P.40: Physically Consistent Analysis of Water Adsorption Kinetics in MOF-303 under Adsorption Cooling Conditions

A. Abuelmaaty | King Fahd university for petroleum and menials (SAU)